
When Modalities Disagree: Detecting Strategic Information Asymmetry in Real Estate Through Cross-Modal Alignment

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Abstract

Multimodal fusion treats all inputs as honest observations of reality, but in many domains some modalities are generated by agents with incentives to mislead. We show that zero-shot prompted scoring and supervised scoring measure fundamentally different constructs: supervised models learn what something is *worth*, while prompts capture how hard an agent *tries* to make it look good. Their divergence detects strategic intent, and no price-based supervised objective can recover the signal because the training target reflects value, not effort. We validate this insight in real estate, where listing text and photographs are curated by a seller’s agent while Street View imagery, satellite photos, and public records are not. Our method, **quality-calibrated alignment**, scores each modality on a quality axis via CLIP zero-shot classification and compares the strategic average to the objective average, requiring no labeled data or fine-tuning. On 6,893 sold properties across seven Boston-area municipalities, misaligned listings are half as likely to sell above asking (logistic odds ratio 0.495 for selling above list price, $p = 0.0003$). A mixture-of-experts model trained only to predict sale price independently learns to discount strategic modalities for misaligned properties: zeroing out strategic inputs shifts log-price predictions by +0.093 for misaligned listings (roughly +10% in price; removing the misleading signal helps) but by −0.073 for aligned ones (roughly −7%; removing truthful signal hurts), with $p < 0.0001$. We formalize this as **adversarial modality interactions**, extending the PID interaction taxonomy [9] to decompose strategically generated unique information into truthful and adversarial subcomponents.

1 Introduction

A listing agent describes a ground-floor apartment as a “sun-drenched retreat with vibrant neighborhood energy.” A buyer who checks Google Street View sees a north-facing unit beside a parking structure. Neither source alone produces this insight. The text offers no basis for skepticism; the image provides no claim to evaluate. Only their juxtaposition reveals that the listing is doing something strategic. The structure parallels multimodal sarcasm detection [2], where a transcript and a video each appear sincere but their combination reveals sarcasm. In our setting, the incongruence reflects a seller’s communicative strategy rather than a speaker’s rhetorical intent.

The same problem arises in product reviews, political advertising, and medical self-reports: one data source is shaped by what the generator wants the audience to believe. Standard multimodal fusion treats all inputs as equally reliable. The multimodal interaction literature [7, 9] decomposes information into redundancy, uniqueness, and synergy, but treats all unique information as legitimate. When one modality is strategically generated, its unique information mixes truthful content with adversarial distortion. The existing taxonomy has no category for this. Unlike adversarial robustness, which assumes perturbations to existing inputs, our setting involves modalities *generated* with strategic intent.

Real estate provides an unusually clean testbed because its six data sources split along incentive lines. **Strategic modalities** (listing text, listing photographs) are curated by the seller’s agent. **Objective modalities** (Google Street View, satellite imagery, tabular attributes, census demographics) exist independent of the transaction. Levitt and Syverson [6] document that real estate agents exploit informational advantages through marketing duration, keeping their own homes on market 9.5 days longer and selling for 3.7% more than when representing clients. We identify a distinct channel: presentation quality. The mechanism isn’t fabrication but selective emphasis. Agents devote vocabulary and camera angles to the renovated kitchen while saying nothing about the neighborhood that Street View would reveal.

We propose **quality-calibrated alignment**, a score measuring whether strategic and objective modalities imply the same quality level for a property. The method uses CLIP [10] zero-shot classification with modality-specific prompt pairs. We make three contributions:

(1) This score predicts which properties sell above asking across 6,893 transactions spanning seven municipalities (Pearson $r = 0.089$ between the misalignment score and the fractional price gap, logistic odds ratio 0.495, $p = 0.0003$).

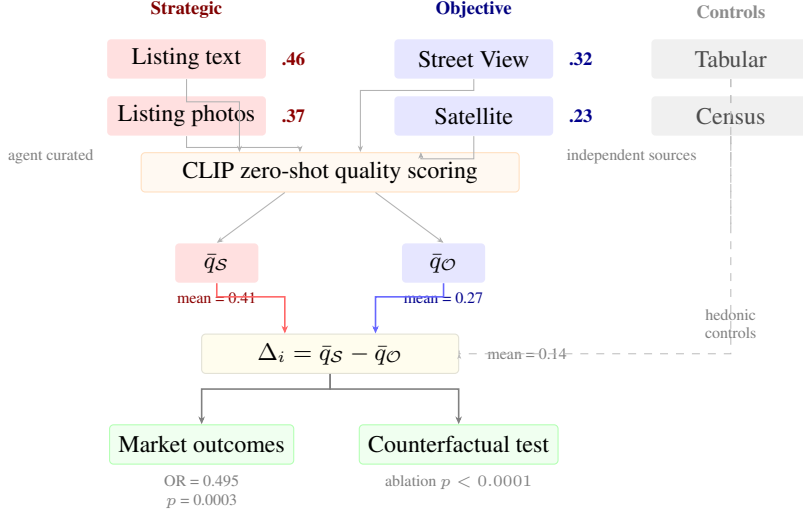


Figure 1: Overview. Strategic modalities (agent-curated, warm) and objective modalities (independent, cool) are each scored on a quality axis via CLIP zero-shot classification. Quality score means (badges) show strategic modalities score systematically higher. The gap Δ_i predicts market outcomes and is independently validated by neural models that learn to discount strategic inputs for misaligned properties.

(2) Neural fusion models trained with no knowledge of our framework independently learn to distrust strategic modalities for misaligned properties (counterfactual ablation, $p < 0.0001$). Early fusion outperforms late fusion by $+0.248$ in cross-validated R^2 for predicting sale price, and the resulting models reveal that cross-modal interactions carry information far beyond what any individual modality provides.

(3) We propose **adversarial modality interactions** as a new category in the PID interaction taxonomy [9], where strategically generated unique information decomposes into truthful and adversarial subcomponents.

The methodological insight with the broadest reach: we tested three supervised alternatives (Ridge, calibrated MLPs, contrastive InfoNCE) on the same CLIP embeddings. All produce misalignment scores with the wrong sign or no signal at all. Supervised models find the *value axis*: what properties are worth. Prompted scoring finds the *effort axis*: how aggressively the agent presents the property. A listing that describes a modest ranch house as “an entertainer’s dream with resort-style living” scores high on the effort axis but low on the value axis. The gap between these two scores is what we measure.

2 Related Work

Multimodal interactions. Liang et al. [7] characterize multimodal information through redundancy, uniqueness, and synergy. Liang et al. [9] develop scalable PID estimators, and Yu et al. [12] propose mixture-of-experts for heterogeneous interaction types. Our setting adds an absent category: strategically generated unique information requires decomposition into truthful and adversarial subcomponents (Section 4.3).

Contrastive models and fusion. CLIP [10] enables zero-shot classification by scoring embeddings against text prompts; we use this to score modalities on a quality scale without domain-specific training. Tsai et al. [11] introduce cross-modal attention; Zadeh et al. [13] propose tensor fusion via outer products of modality representations. We implement both. Liang et al. [8] separate shared from unique representations via factorized contrastive learning; extending this to a three-way factorization is a direction we outline in Section 7.

Visual features in real estate. Law et al. [5] show Street View and satellite imagery improve hedonic pricing models. Hasan et al. [3] learn joint embeddings from property images, text, and tabular data. Huang et al. [4] survey multimodal methods for real estate appraisal. All treat modalities as neutral information sources. None partitions them by the incentive structure of their generators.

Information asymmetry. Akerlof [1] formalizes how asymmetric information degrades market quality. Levitt and Syverson [6] provide evidence that agents exploit informational advantages in real estate. We measure the asymmetry not between buyer and seller, but between modalities the seller controls and modalities the seller cannot manipulate.

3 Problem Statement

Let property i have strategic modalities $\mathcal{S}_i = \{x_i^{\text{text}}, x_i^{\text{photo}}\}$ and objective modalities $\mathcal{O}_i = \{x_i^{\text{sv}}, x_i^{\text{sat}}, x_i^{\text{tab}}, x_i^{\text{geo}}\}$. A frozen CLIP ViT-B/32 encoder maps the two strategic and two objective visual modalities to embeddings $\phi_m(x_i) \in \mathbb{R}^{512}$. Tabular and census features form hedonic controls $\mathbf{x}_i \in \mathbb{R}^p$ (beds, baths, log sqft, property type indicators, tract-level median income,

education rate). We define the fractional price gap:

$$y_i = \frac{p_i^{\text{list}} - p_i^{\text{sold}}}{p_i^{\text{list}}} \quad (1)$$

Negative y_i means the property sold above asking, indicating competitive bidding. We use y_i for the misalignment analysis; for the fusion and architecture comparisons in Section 6.2, the target is log sale price. One caveat: listing price is itself a strategic variable (agents sometimes underprice to trigger bidding wars), so y_i conflates pricing strategy with presentation strategy. Our score captures the visual and textual channel; the pricing channel is orthogonal and not measured here.

We test whether a quality-calibrated misalignment Δ_i , computed from the cross-modal comparison between $\mathcal{S}_i$ and $\mathcal{O}_i$, predicts y_i beyond hedonic controls:

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \beta_\Delta \Delta_i + \varepsilon_i \quad (2)$$

and whether neural fusion models $f_\theta(\mathcal{S}_i, \mathcal{O}_i, \mathbf{x}_i)$ learn to weight strategic modalities differently for high- versus low- Δ_i properties without access to Δ_i during training.

4 Proposed Approach

4.1 Quality-Calibrated Alignment

Raw CLIP cosine similarity between strategic and objective embeddings doesn’t predict outcomes (Pearson $r = 0.029$, $p = 0.41$; Appendix E) because cosine measures topical overlap, not quality agreement. A listing about kitchen renovations has low similarity with a Street View facade regardless of whether the listing is honest.

We project each modality onto a quality axis instead. For modality m , the quality score is the CLIP zero-shot softmax probability of the high-quality anchor in a modality-specific prompt pair:

$$q_m(i) = \frac{\exp(\tau \cdot \phi_m(x_i)^\top \tilde{\phi}(p_m^{\text{hi}}))}{\exp(\tau \cdot \phi_m(x_i)^\top \tilde{\phi}(p_m^{\text{lo}})) + \exp(\tau \cdot \phi_m(x_i)^\top \tilde{\phi}(p_m^{\text{hi}}))} \quad (3)$$

where $\tilde{\phi}(\cdot)$ is the ℓ_2 -normalized CLIP text encoding and $\tau = 100$ is the logit scale used in CLIP zero-shot classification. Prompts are modality-specific (Appendix F): for listing photos, “a photo of a modest, basic home interior” versus “a photo of a luxurious, high-end home interior”; for Street View, “a run-down, undesirable neighborhood street” versus “an attractive, well-maintained neighborhood street.” Each property’s 15–40 listing photos are individually encoded and mean-pooled into a single 512-dimensional representation; Street View images (four cardinal headings) are pooled identically.

The **quality-calibrated misalignment** is the gap between group averages:

$$\Delta_i = \underbrace{\frac{1}{2}(q_{\text{text}}(i) + q_{\text{photo}}(i))}_{\bar{q}_S: \text{strategic quality}} - \underbrace{\frac{1}{2}(q_{\text{sv}}(i) + q_{\text{sat}}(i))}_{\bar{q}_O: \text{objective quality}} \quad (4)$$

Positive Δ_i means strategic modalities imply higher quality than objective ones support. For the 773 listings without photographs, $\bar{q}_S$ uses text alone. The weighting is stable across alternatives: photo-only misalignment ($r = 0.102$), equal weighting ($r = 0.089$), and text-only ($r = 0.076$) all predict price gap at $p < 0.0001$ (Appendix G).

To validate that the quality scores track what we claim, we correlate them with log price per square foot. Strategic quality yields Pearson $r = 0.230$ ($p < 0.0001$); objective quality yields $r = 0.049$ ($p = 0.0001$). CLIP reads quality from curated presentations almost five times more strongly than from uncurated street-level imagery. Linear probes on the raw embeddings show this isn’t because objective modalities lack quality information: all four achieve $R^2 = 0.31$ – 0.33 for predicting residualized price per square foot (Appendix R). The zero-shot prompt direction aligns with economic value for curated content but barely registers for uncurated content. This asymmetry is why prompted scoring captures strategic effort rather than objective value.

4.2 Why Prompted Scoring Beats Learned Scoring

Supervised quality scoring produces misalignment with the wrong sign. We tested three learned alternatives on the same CLIP embeddings, each trained to predict what properties are actually worth:

A Ridge model fit from raw 512-dimensional embeddings to residualized log price per square foot defines a “what the market values” direction. Misalignment computed along this axis correlates $r = -0.056$ with price gap, in the wrong direction. A calibrated MLP (512→256→1, 5-fold CV) produces $r = -0.042$, again the wrong direction. A contrastive model (InfoNCE loss, 128-dimensional shared space) yields $r = 0.003$, indistinguishable from zero. None of the three even produces the correct sign, despite achieving $R^2 \approx 0.27$ for predicting price per square foot (Appendix N).

These three methods all find the value axis: what properties are worth on the market. Prompted scoring finds the effort axis: how aggressively the agent presents the property. For aligned listings, effort and value point in the same direction, consistent with accurate description. For misaligned listings, the effort axis points higher: the agent works harder to dress up a mediocre property. No price-based supervised objective can recover this divergence because the training target reflects value, not effort. The prompts work precisely because they aren’t optimized for prediction; they measure a different construct entirely. Cross-validated misalignment (quality score weights learned out-of-fold on price gap) exceeds the hand-crafted signal ($r = 0.147$ vs. 0.106), confirming the direction is learnable even though the optimal weighting can be refined.

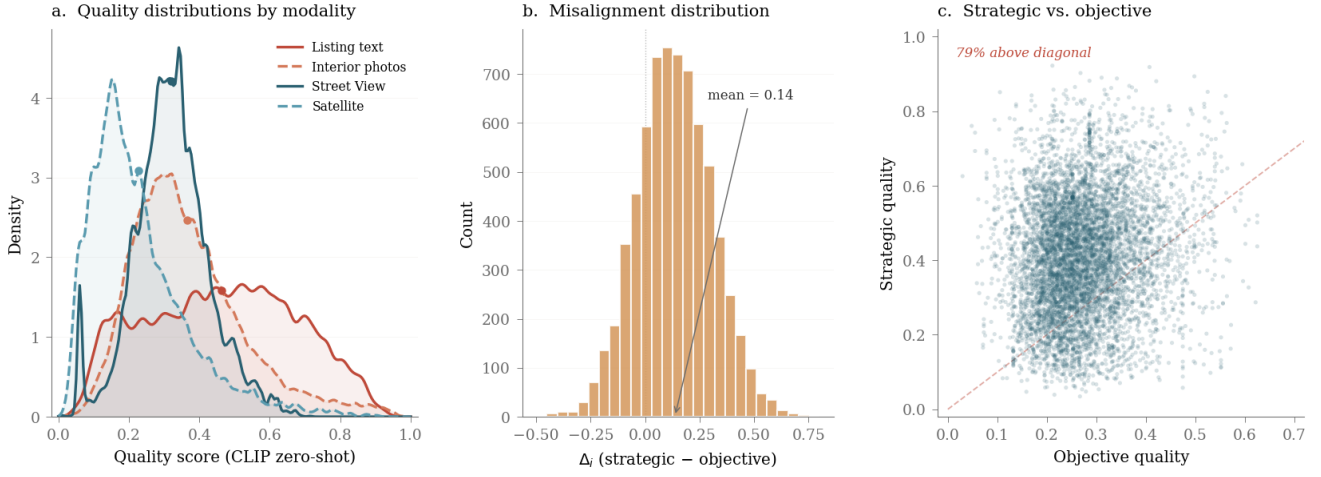


Figure 2: **(a)** Quality score densities by modality. Strategic modalities (warm) score higher on the zero-to-one CLIP quality scale; dots mark means. **(b)** Distribution of Δ_i (mean = 0.14). **(c)** Strategic vs. objective quality; agents systematically present properties as higher quality than independent sources support.

Table 1: Dataset summary ($N = 6,893$). Listing photos are curated by the agent; Street View and satellite imagery are collected independently.

Modality	Source	Class	Coverage
Listing text	Zillow	Strategic	6,880 / 6,893
Listing photographs	Zillow CDN	Strategic	6,120 / 6,893
Street View (4 headings)	Google API	Objective	6,893 / 6,893
Satellite tile	Google Maps	Objective	6,893 / 6,893
Tabular attributes	Zillow	Objective	6,893 / 6,893
Census demographics	Census Bureau	Objective	6,889 / 6,893

4.3 Adversarial Modality Interactions

The PID framework [9] decomposes multimodal information about target Y into redundancy R , uniqueness U_S and U_O , and synergy S :

$$I(X_S, X_O; Y) = R + U_S + U_O + S \quad (5)$$

When X_S is strategically generated, its unique component conflates two subcomponents. We propose decomposing it as:

$$U_S = U_S^{\text{rel}} + U_S^{\text{adv}} \quad (6)$$

U_S^{rel} captures truthful private information (a new HVAC system, a completed renovation) that objective modalities can’t observe. U_S^{adv} captures strategic distortion: quality claims the objective modalities contradict. With Y^* as latent true quality, we define:

$$U_S^{\text{adv}} = I(X_S; Y | X_O) - I(X_S; Y^* | X_O) \quad (7)$$

Y^* is unobserved, but the counterfactual ablation in Section 6.2 provides evidence consistent with this decomposition: the neural model’s differential reliance on strategic modalities for aligned versus misaligned properties reflects the adversarial subcomponent without requiring ground-truth quality labels. An approximate PID computed separately per misalignment quartile shows a suggestive pattern: objective uniqueness (U_O) rises from Q1 to Q4, consistent with independent sources carrying more unique information as strategic distortion increases (Appendix P).

Figure 2 shows the empirical basis: strategic modalities systematically score higher on the prompted quality axis.

5 Experimental Methodology

Dataset. We assembled a six-modality dataset of 6,893 sold residential properties in the greater Boston area (Table 1), spanning seven municipalities: Boston (1,300), Newton (1,005), Arlington (981), Medford (979), Cambridge (977), Brookline (965), and Somerville (680). Median sale price is \$1,050,000. Of these, 6,109 have all four CLIP-encodable modalities, and 6,006 have valid price gaps after excluding 198 unsold properties (whose recorded “price” is an asking price, not a market-clearing valuation) and 50 outliers with price gaps exceeding $\pm 50\%$ (Appendix C).

Evaluation. All R^2 values use 5-fold cross-validation. PCA reduction (30 components per modality) and standardization are fit within training folds. Ridge regularization is selected by nested CV ($\alpha \in [10^{-2}, 10^4]$, 20 log-spaced values). Seven PyTorch neural architectures are trained with AdamW, cosine annealing, early stopping (patience 20), and gradient clipping (max norm 1.0).

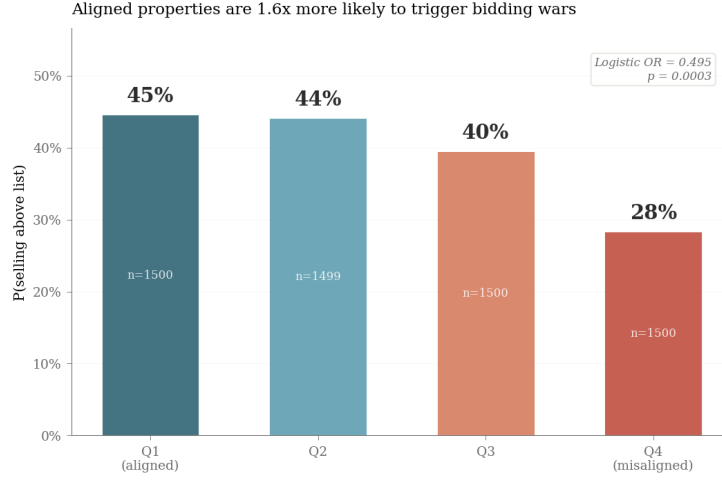


Figure 3: Fraction of properties selling above asking, by misalignment quartile. Q1 (most aligned) vs. Q4 (most misaligned): $1.6\times$ ratio. Logistic OR = 0.495 ($p = 0.0003$) with tabular controls.

6 Results

6.1 Misalignment Predicts Market Outcomes

When multiple buyers submit competing offers for a property, the sale price gets bid above the listed asking price. Properties in the lowest misalignment quartile trigger these bidding wars 44.6% of the time. In the highest quartile: 28.3% (Figure 3). The bivariate Pearson correlation between Δ_i and the fractional price gap is $r = 0.089$ ($p < 0.0001$). A logistic regression with tabular controls yields an odds ratio of 0.495 ($p = 0.0003$): each standard deviation of additional misalignment roughly halves the odds of selling above asking. Bootstrap (2,000 draws): 93.8% of resampled coefficients are positive, 95% CI = $[-0.003, 0.032]$. The signal survives controlling for log days on market ($p = 0.005$), ruling out the possibility that misalignment proxies for how long a listing sits. High-misalignment listings also deploy luxury vocabulary (“stunning,” “exquisite,” “designer”) at $3.1\times$ the rate of low-misalignment listings ($p < 0.0001$; Appendix S), consistent with strategic presentation effort. Full regression specifications appear in Appendix H.

The effect varies with property type. Single-family homes, where every property has a unique exterior, show the strongest signal ($r = 0.131$, $p < 0.0001$; bidding war rates of 53% vs. 43% for aligned vs. misaligned halves). Condos are weaker ($r = 0.063$, $p = 0.001$) because units in the same building share Street View imagery, compressing objective quality variance. Multi-family investment properties show a *reversed* effect ($r = -0.074$, $p = 0.057$): investment buyers evaluate properties through discounted cash flow on projected rental income, and visual presentation matters far less than cap rate and vacancy rate. Aggressive marketing for a multi-family may signal the seller’s confidence in rent projections, which is actually informative. Townhouses track single-family homes ($r = 0.127$, $p = 0.002$; Appendix J). Geographically, the signal is strongest where properties are most differentiated (Brookline $r = 0.145$, Cambridge $r = 0.121$, Newton $r = 0.109$) and weakest where housing stock is most homogeneous: Boston proper ($r = -0.011$), dominated by large condo developments with developer-provided stock photography that leaves little room for individual agent curation (Appendix K). At the brokerage level, mean misalignment correlates $r = -0.351$ ($p < 0.0001$) with the bidding war rate: firms with more aspirational presentations achieve worse outcomes for their clients (Appendix X).

6.2 Fusion Models Learn to Discount Strategic Inputs

Table 2 compares fusion strategies and neural architectures for predicting log sale price. Late fusion (averaging five independent Ridge predictions) achieves $R^2 = 0.503$. Early fusion (concatenating all modality features into a single Ridge) reaches $R^2 = 0.752$, a gap of $+0.248$. This gap quantifies how much predictive information lives in cross-modal feature interactions rather than in individual modality predictions. Polynomial Ridge with 25 explicit cross-group interaction terms closes only 0.002 of this gap, meaning the interactions aren’t simple pairwise products. The neural models close more: MMoE [12] and a cross-modal transformer inspired by Tsai et al. [11] both reach $R^2 \approx 0.81$ (the 0.001 difference is within fold variance), beating Ridge by $+0.063$ in R^2 . Augmented tensor fusion [13] reaches 0.794. Shapley decomposition assigns 34.7% of modality value to photographs, consistent with photos being the primary carrier of both quality signal and strategic distortion (Appendix L).

Every modality’s incremental R^2 over the tabular baseline roughly doubles for misaligned versus aligned properties (e.g., photos: $+0.102$ vs. $+0.057$; Appendix Q). Additional data sources become most valuable precisely when the existing information is unreliable.

Counterfactual modality ablation. For each test property, we compared the cross-modal transformer’s full prediction against its prediction with strategic modalities zeroed out. The shift $\hat{y}_{\text{full}} - \hat{y}_{\text{no-strat}}$ (in log-price units) measures reliance on strategic information. For aligned properties (below-median Δ_i), the mean shift is -0.073 (about -7% in price): removing

Table 2: Predicting log sale price ($n = 6,109$, 5-fold CV R^2).

Architecture	Params	R^2	$\pm$ s.d.
MMoE (6 experts) [12]	45,426	0.815	0.039
Cross-modal transformer [11]	96,129	0.814	0.044
Augmented tensor fusion [13]	25,313	0.794	0.047
MLP (64 hidden)	8,385	0.770	0.033
Ridge (early fusion)	130	0.752	0.036
Late fusion (5 Ridge avg)	130	0.503	—

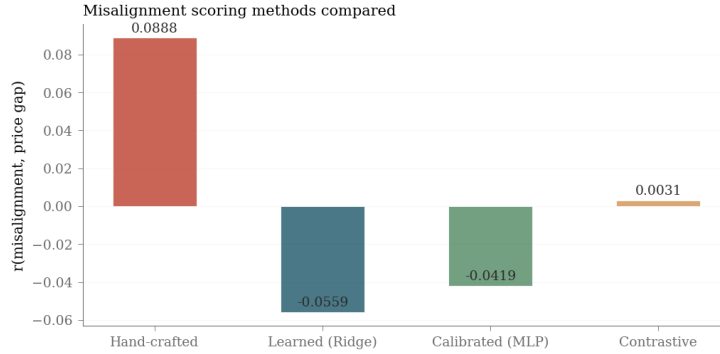


Figure 4: Pearson r between each misalignment measure and the fractional price gap. Only prompted scoring (first bar) produces the correct sign. Supervised methods find the value axis ($r < 0$); prompts find the effort axis ($r > 0$). See Section 4.2.

strategic inputs hurts, because the information is truthful. For misaligned properties, the mean shift is $+0.093$ (about $+10\%$): removing strategic inputs *helps*, because the information misleads. The difference is significant at $p < 0.0001$. A model trained to minimize prediction error, with no access to Δ_i or any knowledge of our incentive partition, independently discovered that strategic modalities are reliable for some properties and adversarial for others. This result is consistent with the U_S^{adv} decomposition in Eq. 6.

Gate and expert structure. The gated fusion model uses photos as a diagnostic channel: its photo gate weight correlates $r = 0.231$ ($p < 0.0001$) with Δ_i . For misaligned properties, the photo embedding encodes the discrepancy between curated presentation and objective reality, and the model learns to attend to this discrepancy signal rather than being misled by it. MMoE specializes experts by property type without supervision: expert 0 handles single-family homes (mean price \$1.65M), expert 2 handles condos (\$955K). Full interpretability results appear in Appendix M.

7 Next Steps

Two limitations bound the current findings: the analysis is observational (no random assignment of misalignment), and the method depends on hand-crafted prompts whose sensitivity to wording we haven’t systematically tested. The following extensions would sharpen the interpretation.

Causal identification. The 64 addresses appearing with different brokers offer within-property variation. For the same physical property, did misalignment change with the broker, and did outcomes follow? One case supports the hypothesis: 17 Berkeley St was listed at $\Delta = 0.299$ and $\Delta = 0.025$ by different brokers; the high-misalignment listing sold 15% below asking, the low-misalignment listing sold 12% above. A larger sample of broker-switchers would make this analysis definitive.

Three-way information factorization. Eq. 6 is a proposed decomposition, not a learned one. One path toward estimating it: extend FactorCL [8] with a three-way contrastive objective using cross-modal consistency as supervision. The counterfactual ablation provides a training signal: properties where removing strategic modalities improves prediction are those with high U_S^{adv} .

Prompt sensitivity and cross-domain transfer. Generating alternative prompt pairs and measuring signal stability would separate robust findings from artifacts of specific word choices. More broadly, the effort-versus-value distinction applies wherever agents generate content strategically: product reviews, political messaging, job applications. We predict that zero-shot quality scoring will outperform supervised scoring for detecting strategic intent in any of these domains.

Buyer heterogeneity. The multi-family reversal suggests buyer population moderates the effect. Testing whether investor-versus-owner-occupant composition predicts heterogeneous treatment effects would strengthen the causal interpretation and connect to the broader question of when strategic modalities mislead versus inform.

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A Group Work

Stephanie Chen designed and built the data collection infrastructure for this project. Stephanie wrote the Zillow scraping pipeline using the Hasdata API, implemented Google Street View and satellite image downloads, set up Census tract demographic retrieval, built the listing detail extraction system including price history parsing, wrote the data cleaning workflow, and ran the collection pipeline across multiple municipalities. **Sophie Lin** researched data sourcing strategies and API options, brainstormed the collection approach together with Stephanie, ran the pipeline across multiple municipalities, managed image storage and file organization, performed quality assurance on the collected data (checking for missing images, parsing errors, and geographic coverage gaps), and together with Stephanie, handled the expansion from the original single-city dataset to the final seven-city coverage. Together, Stephanie and Sophie produced the six-modality dataset of 6,893 properties that this project depends on. Stephanie and Sophie also conducted the initial literature review, and all three authors collaborated on brainstorming and writing the original project proposal.

Joseph Firmansyah identified real estate as a domain where multimodal data splits cleanly along incentive lines (listing text and photos are agent-curated; Street View, satellite, and public records are not) and formulated the research question: does cross-modal quality alignment between these groups reveal strategic behavior, and can the market detect it? Joseph developed the quality-calibrated alignment method, devised the comparison between prompted and supervised scoring as a way to distinguish presentation effort from actual value, coined the term *adversarial modality interactions* and defined the accompanying framework extending PID, designed the experimental pipeline (statistical robustness, neural fusion architectures, interpretability, contrastive and calibration baselines), conducted the statistical and modeling analyses, produced the figures, interpreted the results, and wrote this paper.

B Code Repository

Code, the analysis notebook, and all figures are available at <https://github.com/joseph-mit/MultiModalAI>.

C Data Collection and Cleaning

We built a custom Python pipeline to collect residential property data for the greater Boston area, covering Boston, Cambridge, Somerville, Brookline, Arlington, Medford, and Newton. Property listings were scraped from Zillow via the Hasdata API, filtered to sold properties. For each property we collected standard listing attributes (price, beds, baths, square footage, coordinates) along with listing photo URLs.

For each property we downloaded three types of images: exterior and interior listing photos (pulled directly from Zillow’s CDN), four Google Street View images at 0, 90, 180, and 270 headings, and one satellite image from the Google Maps Static API. Census demographic data (median income, home value, rent, racial composition, educational attainment) was retrieved from the 2023 ACS 5-Year Estimates using the property’s coordinates to identify its Census tract. A secondary API call to Hasdata’s property detail endpoint retrieved full listing descriptions and price history, from which we extracted listing price, listing date, sale date, and days on market.

We applied a cleaning script to produce the final dataset. Key steps included parsing photo URL strings into actual lists, coercing numeric columns, and remapping image file paths to local disk locations by checking file existence from property IDs. The Census Bureau encodes missing income as -666666666 ; these were set to missing and median-imputed. We computed a price gap variable $(p_{\text{list}} - p_{\text{sold}})/p_{\text{list}}$, where positive values mean the property sold below asking and negative values mean it sold above. Properties with price gaps exceeding $\pm 50\%$ were flagged as outliers and excluded, as these typically reflect re-listings or scraping errors. Since data collection took many hours across multiple runs, the pipeline was designed to be safely resumable: image downloads skip files that already exist on disk, census lookups save after each property, and CLIP embeddings are checkpointed atomically every 100 properties.

D Dataset Overview

Figure 5 summarizes the dataset. Sale prices range from \$130K to over \$14M, right-skewed toward a luxury tail of Beacon Hill and Seaport condos. Days on market peaks near 30–90 days with a long right tail of properties that required price cuts. Condos make up 48% of the sample, consistent with Boston’s urban housing stock, followed by single-family homes (29%), multi-family (11%), and townhouses (9%).

E Why Raw Cosine Similarity Fails

Figure 6 illustrates why raw CLIP cosine similarity between strategic and objective embeddings doesn’t predict outcomes (Pearson $r = 0.029$, $p = 0.41$). Cosine captures topical content (kitchen interiors vs. building exteriors), not quality agreement. A listing about a kitchen renovation has low cosine similarity with a Street View facade regardless of whether the description is accurate. Strategic-objective pairs show wider variance than objective-objective pairs, reflecting variable levels of agent curation, but this variance doesn’t map onto market outcomes.

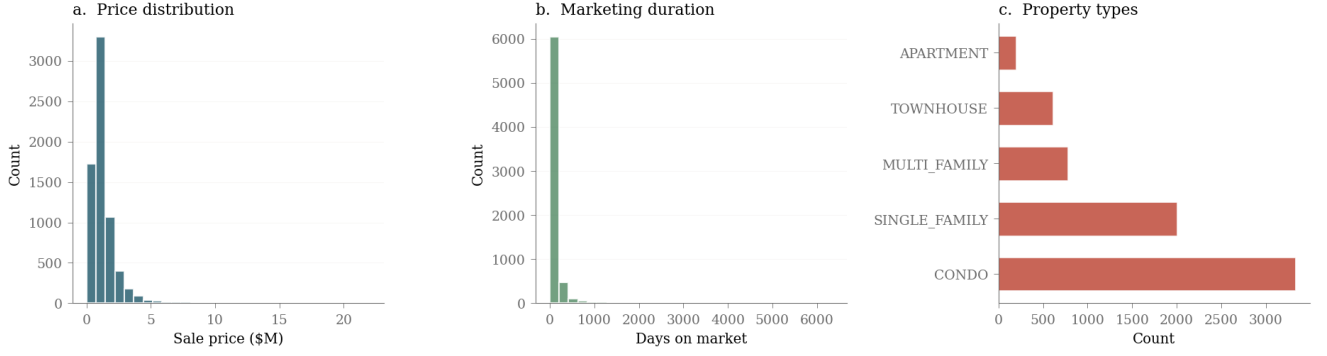


Figure 5: **(a)** Sale price distribution (\$130K–\$14M+). **(b)** Days on market. **(c)** Property type counts across seven municipalities.

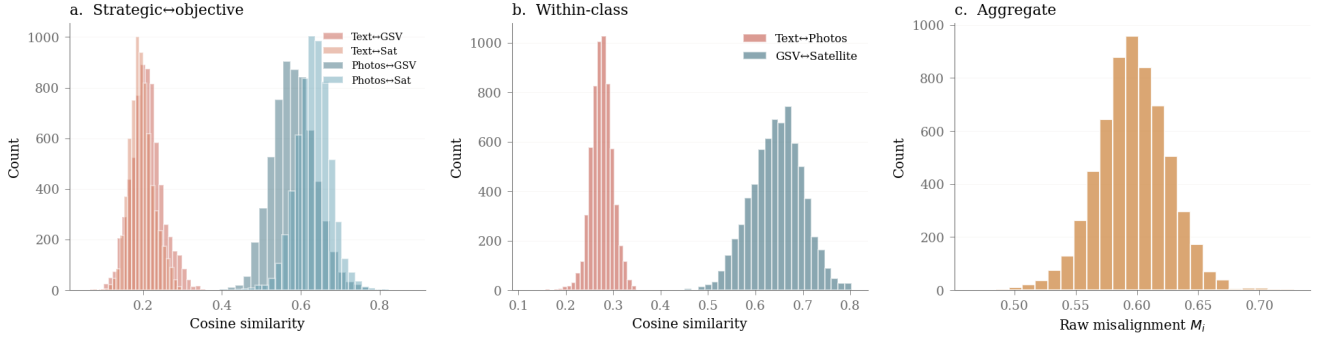


Figure 6: **(a)** Strategic-objective cosine similarities across four modality pairs. **(b)** Within-class similarities: strategic pairs (text↔photos) vs. objective pairs (GSV↔satellite). **(c)** Aggregate raw misalignment, which fails to predict price gap.

F CLIP Prompt Templates

Scope	Low anchor	High anchor
<i>Overall quality</i>		
Text	“listing for a modest, basic property”	“listing for a luxurious, upscale property”
Photos	“photo of a modest, basic home interior”	“photo of a luxurious, high-end home interior”
GSV	“run-down, undesirable neighborhood street”	“attractive, well-maintained neighborhood street”
Satellite	“dense, low-income urban area”	“affluent residential area”
<i>Per-attribute</i>		
Bldg. cond.	“poorly maintained, run-down building”	“well-maintained, pristine building”
Int. quality	“basic, dated interior, cheap finishes”	“luxurious interior, high-end finishes”
Neighborhood	“run-down, undesirable neighborhood”	“beautiful, upscale residential neighborhood”
Space/light	“small, dark, cramped room”	“spacious, bright, airy room”

G Robustness Checks

Table 3 summarizes robustness tests. The main finding holds across every alternative definition and weighting we tested. All pairwise modality gaps predict price gap at $p < 0.0001$. Photos carry the strongest individual signal ($r = 0.102$), consistent with photographs being the primary channel through which agents curate strategic presentation.

The controlled regression with text features (luxury word count, description length, superlative density) pushes the misalignment coefficient to $p = 0.092$. We interpret this as over-controlling: luxury word count is a mediator of strategic behavior, not a confounder. The agent who inflates the visual quality gap *also* uses more luxury language; controlling for the mechanism partitions the effect between channels rather than testing whether it exists. The logistic regression (OR = 0.495, $p = 0.0003$) and DOM mediation ($p = 0.005$) provide cleaner tests.

Table 3: Robustness tests ($n = 5,999$). All r values are Pearson correlations with the fractional price gap.

Category	Test	Result
Statistical	Bootstrap (2,000)	93.8% positive; CI = $[-0.003, 0.032]$ $p = 0.090$
	Permutation (10,000)	
Controls	Tabular only	Δ_i $p = 0.046$
	Tabular + text features	Δ_i $p = 0.092$
	DOM mediation	Δ_i $p = 0.005$
	Within-broker FE	Δ_i $p = 0.063$
Definition	Ratio $\bar{q}_S / \bar{q}_O$	$r = 0.073, p < 0.0001$
	Max pairwise gap	$r = 0.114, p < 0.0001$
Weighting	Photos only	$r = 0.102, p < 0.0001$
	Equal (default)	$r = 0.089, p < 0.0001$
	Text only	$r = 0.076, p < 0.0001$
Learning	Cross-validated Δ	$r = 0.147, p < 0.0001$

H Regression Details

Table 4 shows the progressive regression specifications predicting the fractional price gap. The low R^2 values reflect the nature of the dependent variable: price gap captures divergence between listing price and sale price, driven by pricing strategy, market timing, bidding dynamics, and negotiation. A zero-shot CLIP score explaining any systematic component of this noisy outcome is meaningful despite the modest variance explained.

Table 4: Predicting fractional price gap ($n = 5,999$). Price gap = (list − sold) / list.

Specification	R^2	β_Δ	$p(\beta_\Delta)$
Tabular only	0.052	—	—
+ text features	0.068	—	—
+ misalignment Δ_i	0.053	0.016	0.046
+ text + Δ_i	0.069	0.014	0.092
+ text + Δ_i^{neigh}	0.068	0.005	0.538

Figure 7 shows the continuous relationship. The Pearson correlation with price gap ($r = 0.089$) is modest but consistent across every robustness check. The near-zero correlation with days on market ($r = 0.004$) rules out the possibility that misalignment proxies for marketing duration.

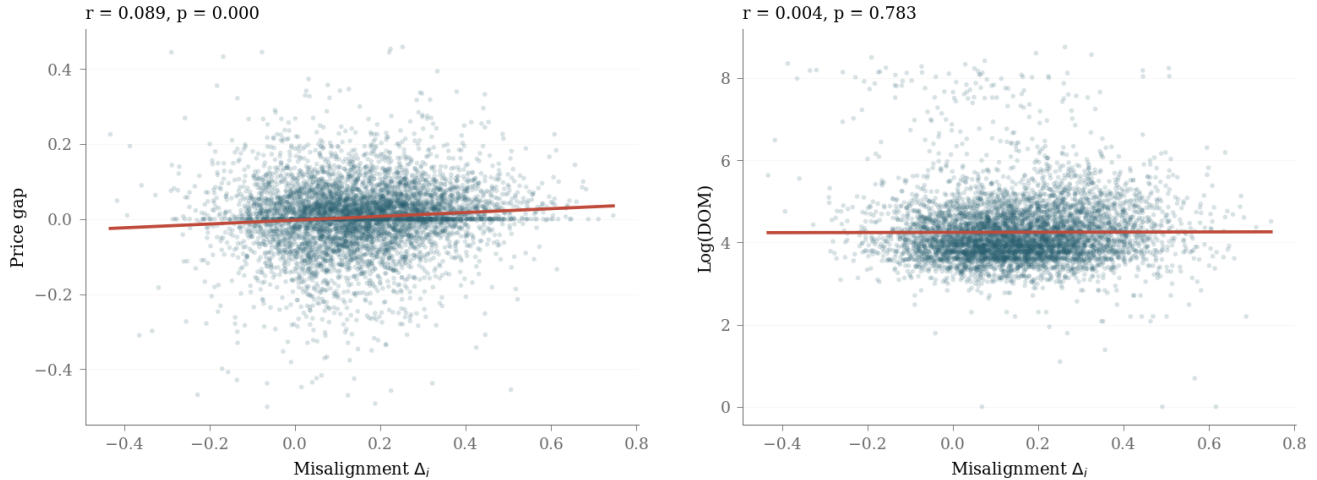


Figure 7: Misalignment Δ_i vs. fractional price gap (left, $r = 0.089$) and log days on market (right, $r = 0.004$). The pricing effect isn’t mediated by how long the property sits.

I Per-Attribute Decomposition

We decomposed Δ_i into four attribute channels using attribute-specific prompt pairs (Appendix F). Neighborhood gap is the strongest predictor of price gap ($r = 0.057, p < 0.0001$): listings that overstate neighborhood quality relative to what

Street View shows achieve worse outcomes. Interior quality is second ($r = 0.038$, $p = 0.005$). Building condition washes out ($r = 0.007$, $p = 0.60$), likely because expanded geographic coverage brought in more properties where condition claims (“new roof,” “gut renovation”) reflect genuine improvements buyers verify at inspection.



Figure 8: Strategic quality (red) systematically exceeds objective quality (teal) across all four attributes. Gap annotations show the difference; stars indicate the gap’s correlation with price gap is significant.

J Property Type Analysis

Single-family homes show the strongest signal because each has a unique exterior and the owner curates the listing. Condos are weaker because units in the same building share Street View imagery, compressing objective quality variance. The multi-family reversal ($r = -0.074$) is consistent with a different buyer population: investors who price off rental income cap rates rather than visual presentation. The signal is also stronger above the city-specific median price ($r = 0.144$, $p < 0.0001$) than below it ($r = 0.032$, $p = 0.081$).

Table 5: Misalignment effect by property type. BW: fraction selling above asking for low- vs. high- Δ_i halves.

Type	n	$r(\Delta, y)$	p	BW (low Δ)	BW (high Δ)
Single-family	1,876	0.131	<0.0001	0.531	0.433
Townhouse	569	0.127	0.002	0.400	0.271
Apartment	85	0.302	0.005	0.395	0.214
Condo	2,797	0.063	0.001	0.382	0.298
Multi-family	672	-0.074	0.057	0.393	0.429

K City-Level Analysis

The signal concentrates in wealthier suburban cities with differentiated housing stock (Figure 9). Boston’s null result ($r = -0.011$, 1,300 properties) reflects a condo-dominated market where developer-provided professional photography is standard and individual agents have less room to curate distinctive presentations.

Figure 10 summarizes effect sizes. Dot size encodes sample size; color encodes median price. The positive relationship between city wealth and misalignment effect size suggests the score captures a phenomenon most active in markets where presentation norms are more aspirational.

Of 47 testable subgroups (property type $\times$ city $\times$ price tier), 14 (30%) show a significant positive signal at $p < 0.05$ (Figure 11). The strongest: Newton single-family ($r = 0.220$, $n = 473$), Somerville single-family ($r = 0.439$, $n = 73$), Cambridge townhouses ($r = 0.281$, $n = 110$). Multi-family shows consistent negative or null effects across all cities.

L Fusion and Architecture Details

Early fusion ($R^2 = 0.752$) outperforms late fusion ($R^2 = 0.503$) by +0.248 in cross-validated R^2 (Figure 12). Adding Δ_i as an explicit feature doesn’t improve early fusion (R^2 stays at 0.752): the concatenated feature space already captures whatever Δ_i summarizes. Polynomial Ridge with 25 cross-group interaction terms adds only +0.002. The neural architectures’ +0.063 improvement shows that the interactions are real but require nonlinear modeling.

Shapley values. Photos: 34.7%, text: 23.6%, satellite: 21.1%, GSV: 20.6%. Photographs carry the largest share.

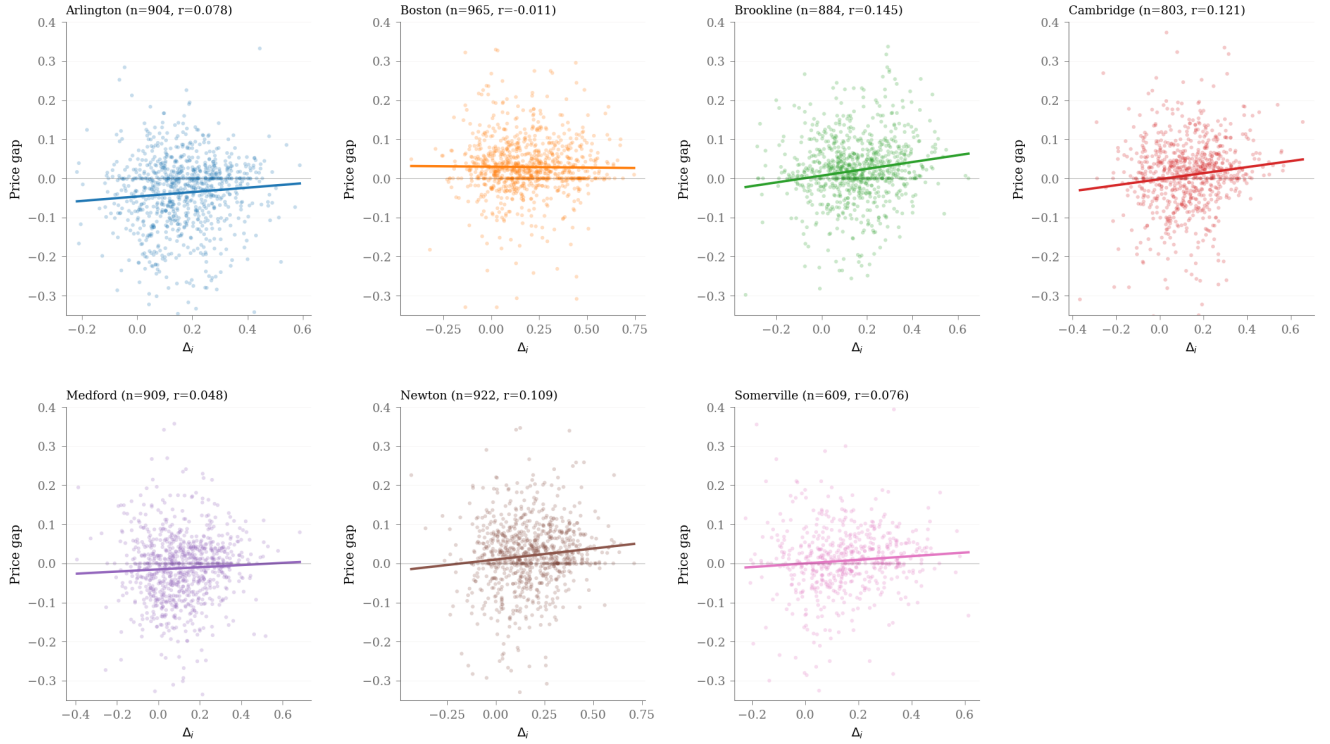


Figure 9: Misalignment vs. fractional price gap by city. Same axis limits across all panels. Brookline ($r = 0.145$), Cambridge ($r = 0.121$), and Newton ($r = 0.109$) show the strongest effects.

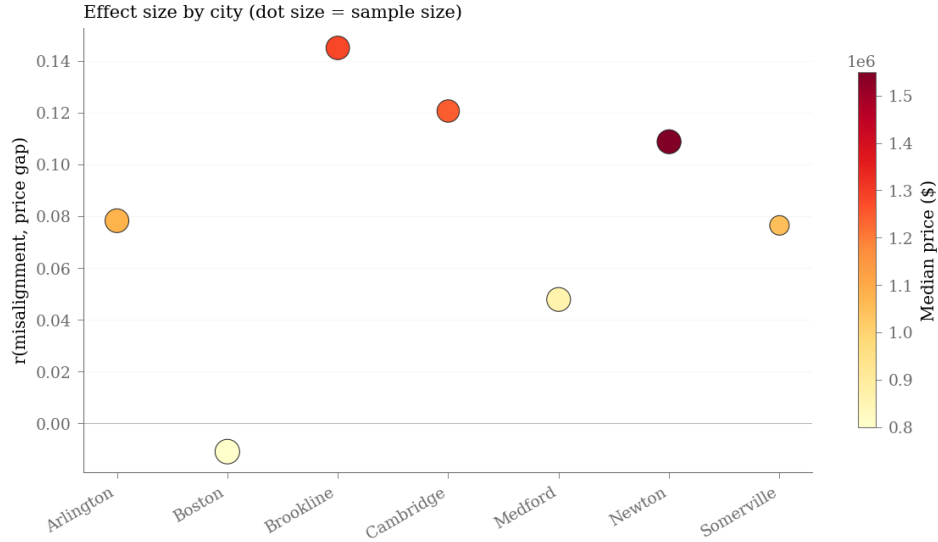


Figure 10: Effect size (r) by city. The signal concentrates where median prices are highest.

Figure 13 shows the complete architecture comparison. Modality masking (zeroing each modality with $p = 0.15$ during training) doesn't help at this sample size. Cross-modal attention produces NaN because the architecture is degenerate for single-vector inputs: with one query and one key per group, the attention weight collapses to a constant.

M Interpretability

Attention heatmaps. Figure 14 shows the cross-modal transformer's attention patterns for aligned versus misaligned properties. Differences are modest but directionally consistent: for misaligned properties, the tabular query attends more to photos (weight 0.435 vs. 0.410) and less to text (0.263 vs. 0.298). The model shifts from verbal to visual evidence when the listing appears suspect.

Gradient importance. Figure 15 shows the mean gradient norm ($\|\partial \hat{y} / \partial x_m\|$) per modality across misalignment quartiles. Photo gradients are largest across all quartiles and increase monotonically from Q1 to Q4, consistent with photos carrying both

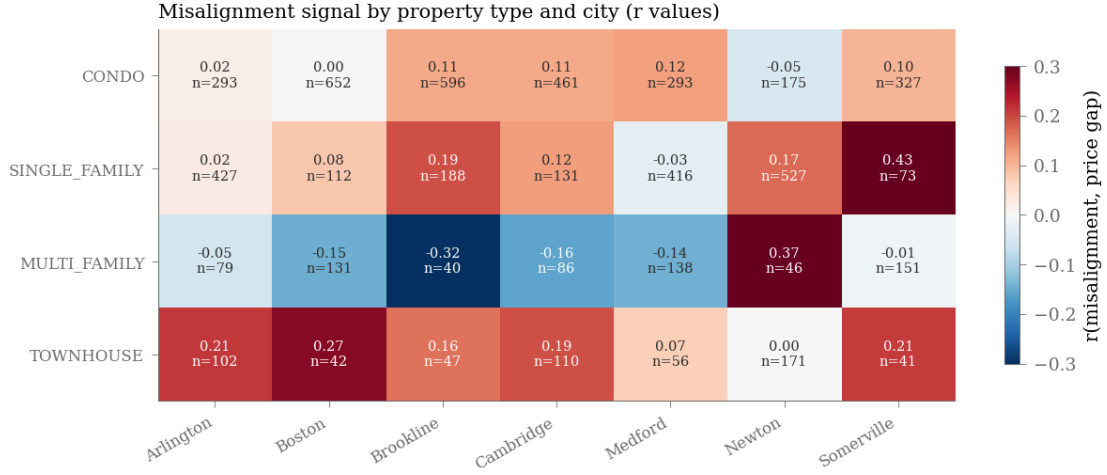


Figure 11: Misalignment signal (Pearson r) by property type and city. Blue: negative. Red: positive. Cells with $n < 15$ greyed out.

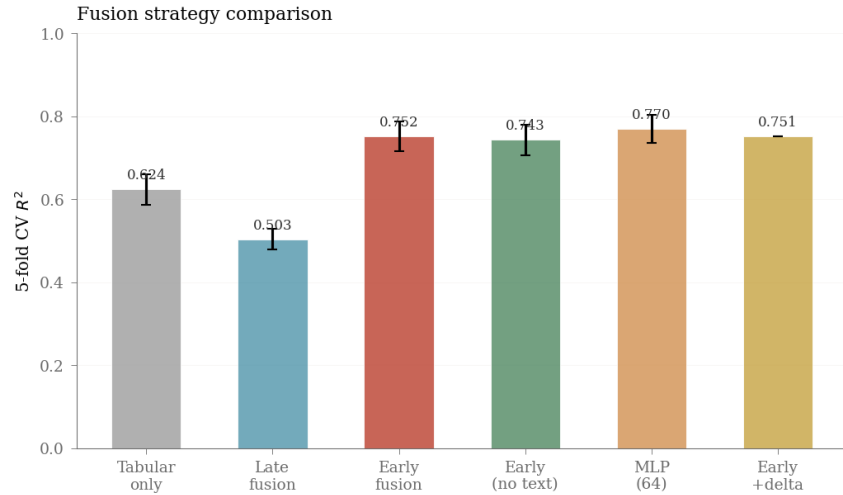


Figure 12: Fusion comparison (5-fold CV R^2 for predicting log sale price). The +0.248 gap between early and late fusion measures the value of cross-modal feature interactions.

the most predictive information and the most strategic distortion. Tabular gradients are smallest and stable: the model treats tabular inputs as a reliable baseline regardless of misalignment.

MMoE expert specialization. The model’s gating network routes different property types to different experts without supervision. Expert 0 handles predominantly single-family homes (mean price \$1.65M); expert 2 handles condos (\$955K). This emergent specialization is consistent with the interaction heterogeneity that Yu et al. [12] designed MMoE to handle.

N Misalignment Scoring Methods

Figure 16 compares four approaches to measuring misalignment from the same CLIP embeddings. The hand-crafted prompted method (Pearson $r = 0.089$ with price gap) is the only one producing a positive, significant correlation. Learned Ridge ($r = -0.056$), calibrated MLP ($r = -0.042$), and contrastive InfoNCE ($r = 0.003$) all fail. The calibration MLPs achieve $R^2 \approx 0.27$ for predicting price per square foot; they find meaningful quality representations, but those representations track what properties are *worth*, not how hard the agent *works* to present them. Section 4.2 explains why this distinction matters.

O CCA Decomposition

Canonical correlation analysis between the 60-dimensional strategic PCA block and the 60-dimensional objective PCA block reveals the shared information structure between modality groups (Figure 17). The top canonical correlation is 0.875, declining to 0.375 by the tenth component. The first five dimensions capture shared information (location, price tier, neighborhood character). The L2 norm of strategic private components (dimensions with canonical correlation below 0.1) correlates $r = 0.066$ ($p = 0.003$) with misalignment: properties whose strategic embeddings carry more private information tend to score higher on Δ_i .

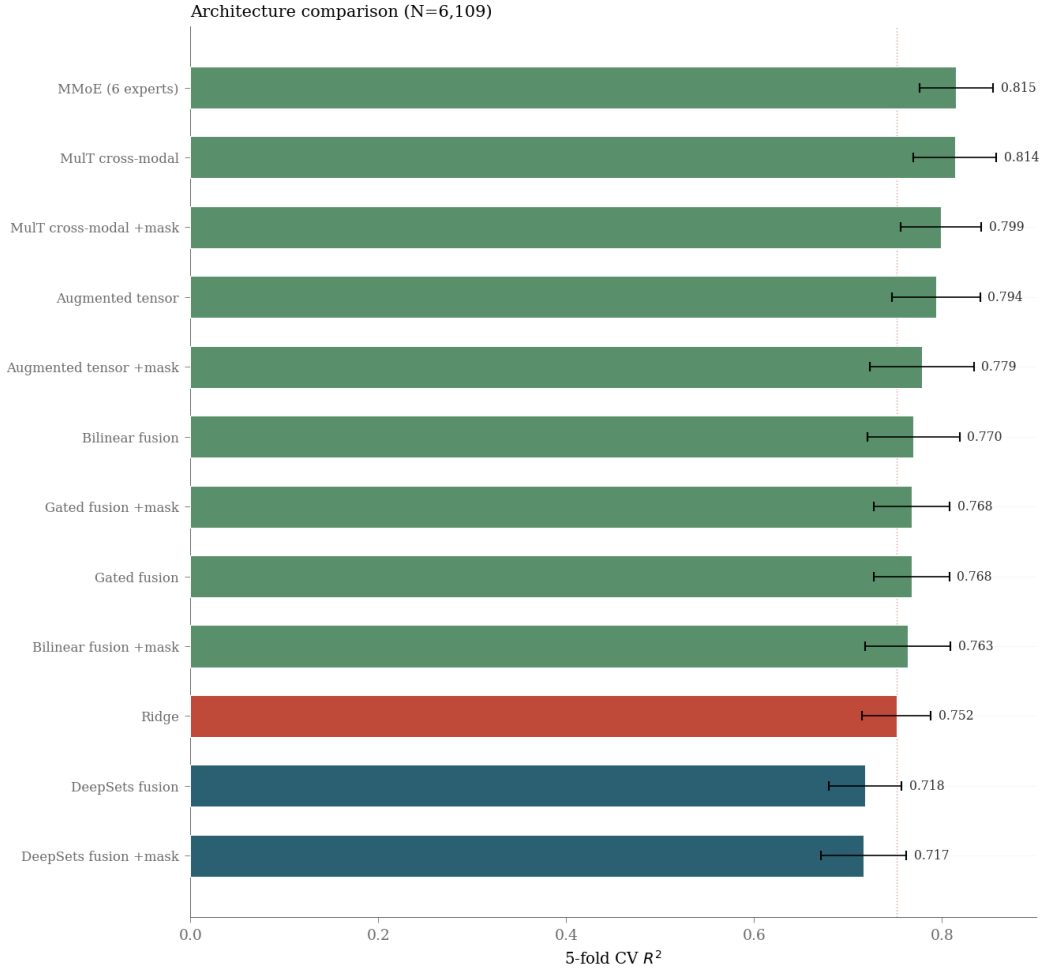


Figure 13: All 14 architecture variants ranked by CV R^2 . Red: Ridge baseline. Green: beats Ridge. Teal: below Ridge.

P PID by Misalignment Quartile

Figure 18 shows an approximate PID (using R^2 as a proxy for mutual information, which does not satisfy the PID axioms but provides directional evidence) computed per misalignment quartile. The numbers are small and should be treated as suggestive. The most consistent pattern: objective uniqueness (U_O) rises from 0.000 in Q1 to 0.010 in Q4. As misalignment increases, objective modalities carry proportionally more information that strategic modalities don't. Strategic uniqueness ($U_S \approx 0.02$) remains elevated across all quartiles.

Q Conditional Modality Value

Figure 19 splits the dataset at the median misalignment and measures each modality's incremental R^2 over a tabular-only baseline for predicting log sale price. For high-misalignment properties, every modality adds value (photos +0.102, GSV +0.087, satellite +0.084, text +0.078). For low-misalignment properties, contributions are roughly halved (photos +0.057, text +0.065, satellite +0.051, GSV +0.047). When the existing information is unreliable, every additional source becomes more valuable. This generalizes: in any multimodal system, additional modalities are most informative when existing sources disagree.

Figure 20 breaks modality value down by property type and price tier. Photographs dominate for condos and above-median properties. Street View contributes most for single-family homes with unique exteriors. Multi-family properties show near-zero or negative visual modality contributions, consistent with investment buyers who base decisions on financial projections.

R Linear Probe Analysis

Linear probes on the raw 512-dimensional CLIP embeddings predict location-residualized log price per square foot at $R^2 = 0.31$ – 0.33 for all four modalities. CLIP encodes quality information for every modality, not just curated content. Probe predictions correlate with zero-shot quality scores for strategic modalities (Pearson $r \approx 0.38$) but weakly for objective modalities ($r \approx 0.15$ for GSV, $r \approx 0.03$ for satellite). The prompted direction aligns with economic value for curated content

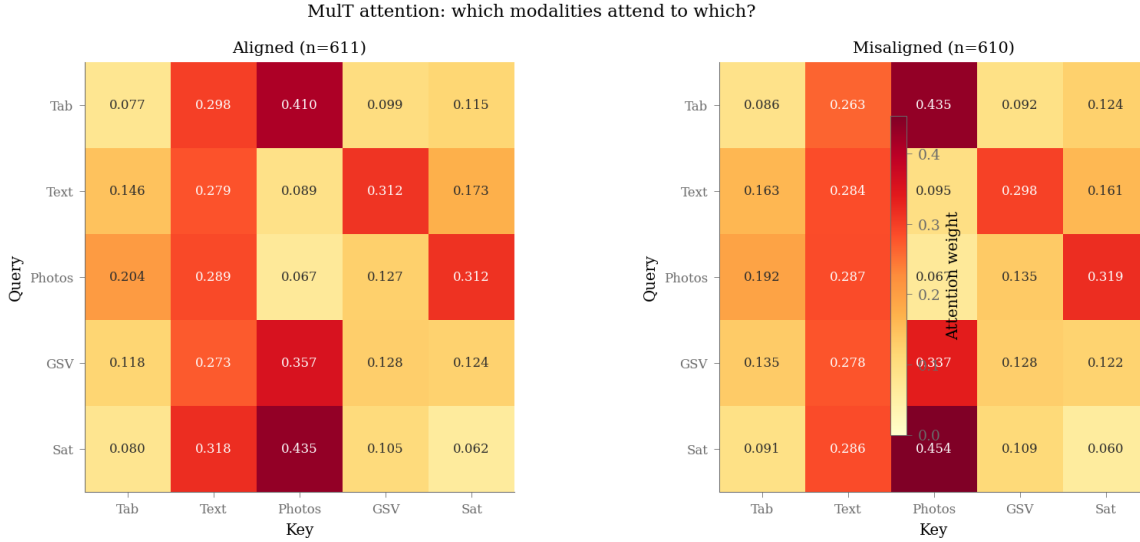


Figure 14: Average attention weights for aligned (left) and misaligned (right) properties. Rows: query modality. Columns: key modality. For misaligned listings, tabular queries shift attention from text to photos.

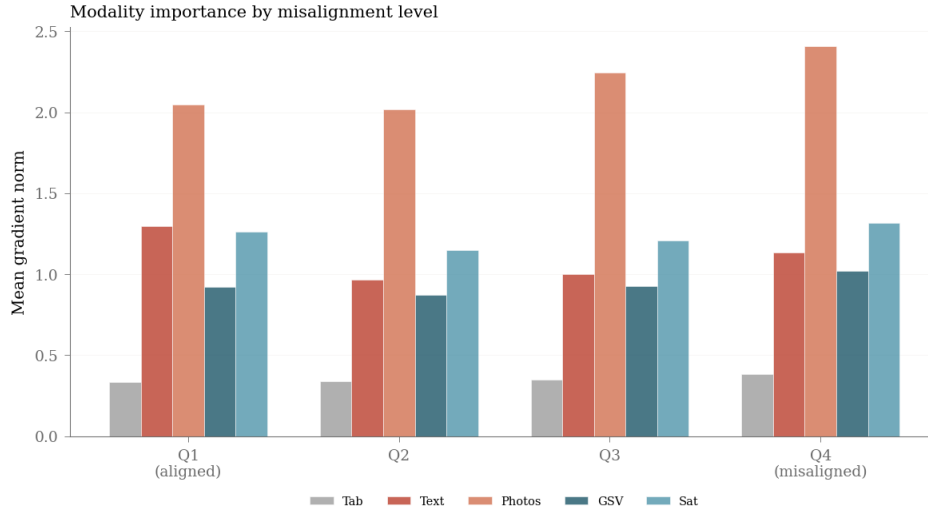


Figure 15: Gradient-based modality importance by misalignment quartile. Photo importance rises with misalignment; tabular remains flat.

but barely registers for uncurated imagery. This asymmetry explains why prompted scoring captures strategic effort: the prompts ask whether content “looks luxurious,” and curated photos of modest properties score high on this axis while uncurated Street View of the same property does not.

S Topic and Text Analysis

Figure 21 compares listing vocabulary between high- and low-misalignment halves. High-misalignment listings deploy luxury vocabulary at $3.1\times$ the rate of low-misalignment listings ($p < 0.0001$). Views and outdoor mentions appear at $1.4\times$. Neighborhood language is elevated more modestly ($1.25\times$). Every topic category runs higher in high-misalignment listings, consistent with agents working harder across the board. High-misalignment listings are also longer on average (903 vs. 769 words, $p < 0.0001$), though exclamation mark usage doesn’t differ ($p = 0.055$).

T Spatial Analysis

Misalignment concentrates in downtown Boston luxury neighborhoods (Back Bay, Beacon Hill, Seaport) and affluent suburbs (Newton, Brookline). The price gap shows a correlated but noisier spatial pattern (Figure 22).

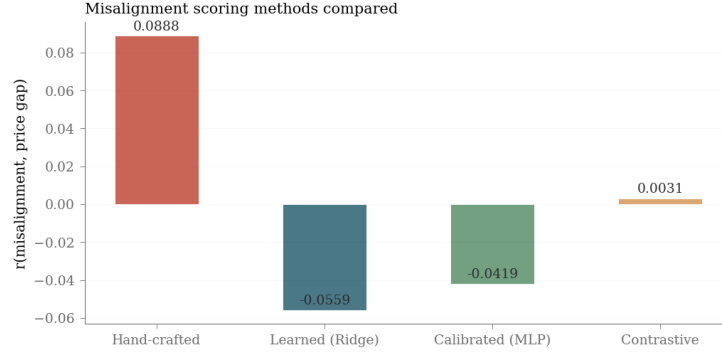


Figure 16: Four misalignment scoring methods, all using the same CLIP embeddings. Only the prompted approach produces the correct sign.

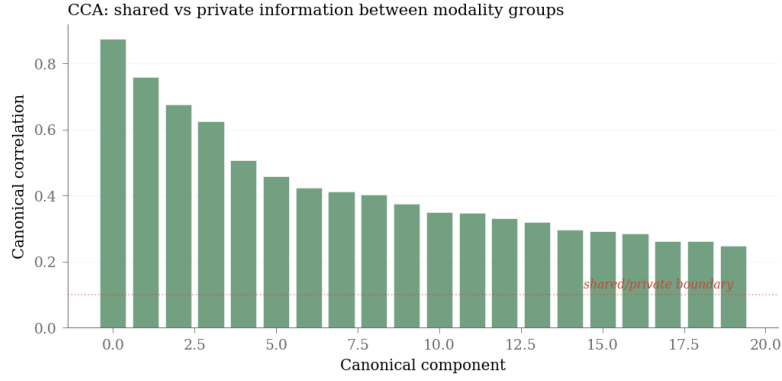


Figure 17: CCA canonical correlations between strategic and objective embedding blocks. Green: shared dimensions (> 0.1). Grey: private dimensions.

U No-Photo Listings

Of the 6,893 properties, 773 have no listing photographs. These sell *better*: 44.1% trigger bidding wars versus 38.9% for listings with photos (Figure 23). Text-only misalignment for no-photo listings correlates $r = -0.105$ ($p = 0.054$) with price gap, the opposite sign from the main finding. Agents who skip photos may be doing so for properties that sell themselves, not properties that need help.

V Cross-Modal Correlation Structure

Strategic modalities show moderate positive within-class correlation; objective modalities show near-zero (Figure 24). Cross-class correlations are weak. Photos are the primary driver of misalignment (Pearson $r = 0.75$ with Δ_i); satellite pulls the objective average down consistently.

W Interaction Types

Following Yu et al. [12], we assigned each property to an interaction type. Synergistic properties (26%) require cross-modal reasoning. The misalignment signal is significant for synergy ($r = 0.086$, $p < 0.001$), unique-S ($r = 0.089$, $p = 0.001$), and unique-O ($r = 0.073$, $p = 0.046$), but not for redundant properties ($r = 0.037$). Misalignment matters most when modalities carry complementary information.

X Broker Analysis

Misalignment varies systematically across brokerages (ANOVA $F = 3.99$, $p < 0.0001$). Luxury specialists show the highest scores: MP Boston Marketing ($\bar{\Delta} = 0.57$, mean price \$3.5M), Campion & Company ($\bar{\Delta} = 0.46$, \$8.5M). At the broker level ($n = 33$ firms with 5+ listings), misalignment correlates $r = -0.351$ ($p < 0.0001$) with the bidding war rate. Aspirational presentation is a luxury-market house style that doesn't translate into better outcomes for sellers.

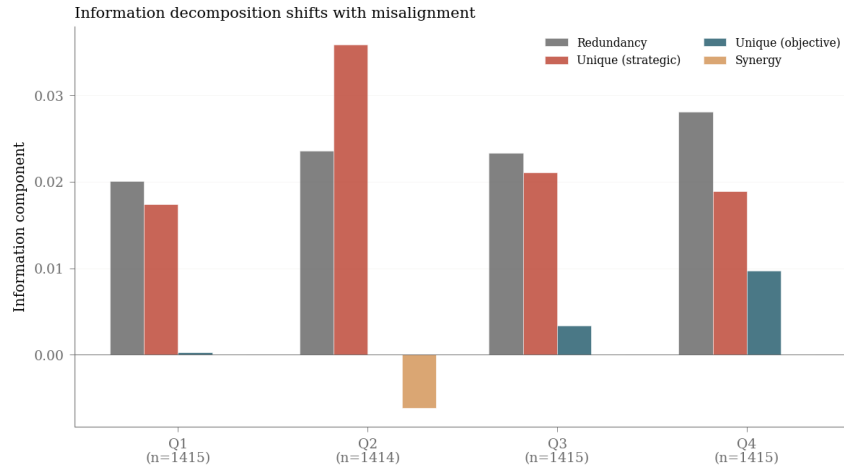


Figure 18: Approximate PID components by misalignment quartile. Objective uniqueness rises with misalignment; strategic uniqueness remains elevated throughout.

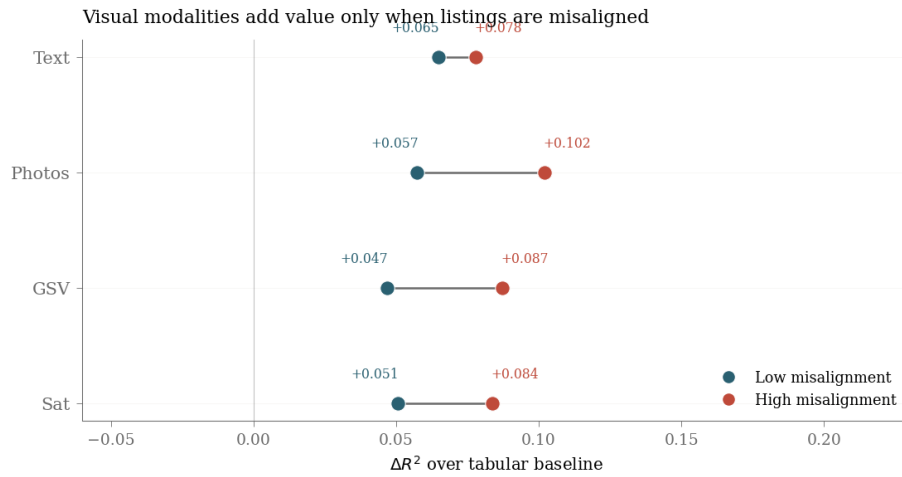


Figure 19: Incremental R^2 per modality over tabular baseline, by misalignment level. Every modality contributes more when misalignment is high.

Y Case Studies

The highest-misalignment property is 1 Franklin St Unit 3907 ($\Delta = 0.75$, \$2.08M), a luxury condo whose listing emphasizes “soaring ceilings” and “designer finishes” while Street View shows a nondescript glass tower. It sold at exactly list price. The lowest is 20 Kinmonth Rd #203 ($\Delta = -0.43$, \$696K), a modestly presented condo that sold 23% above asking. Conservative presentation, strong outcome.

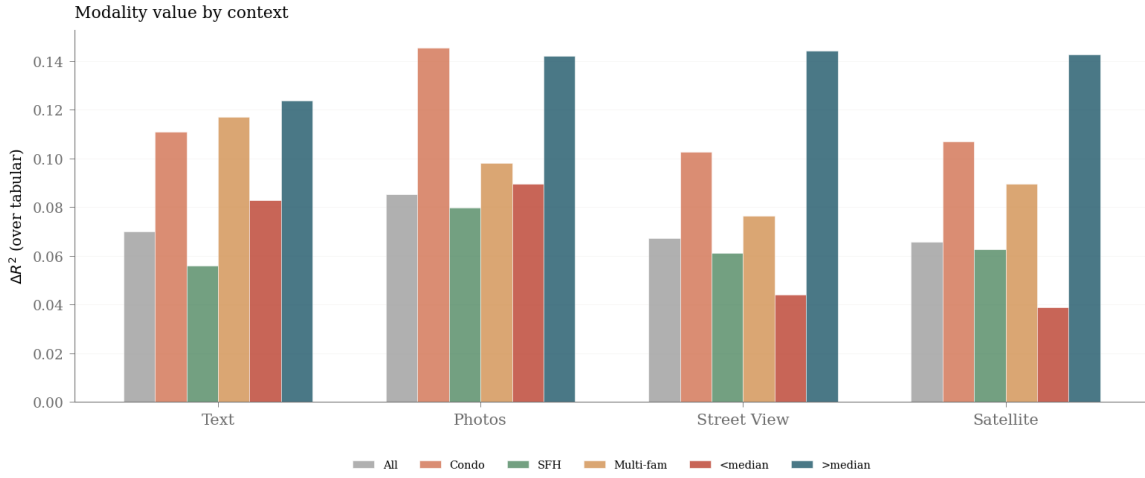


Figure 20: Modality value by property context. Visual modalities add little for multi-family investments.

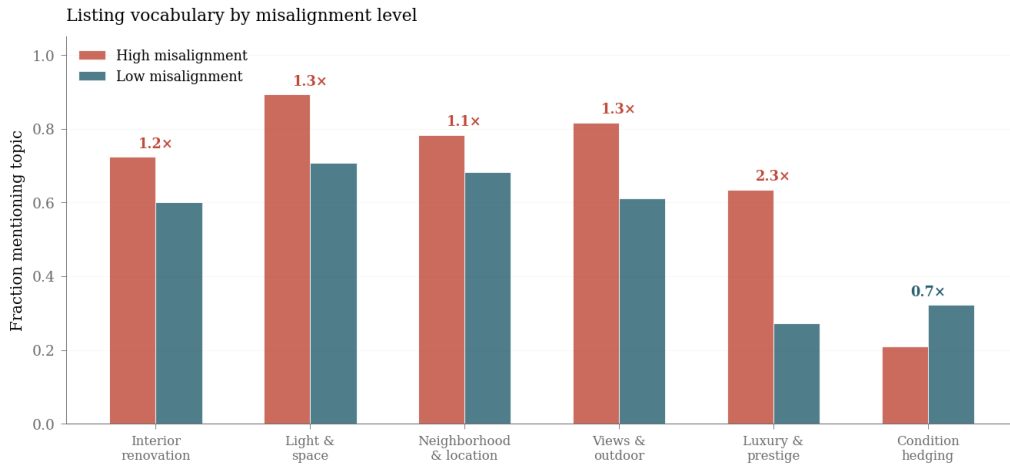


Figure 21: Keyword frequency by misalignment level. High-misalignment agents deploy luxury language at $3.1\times$ the base rate.

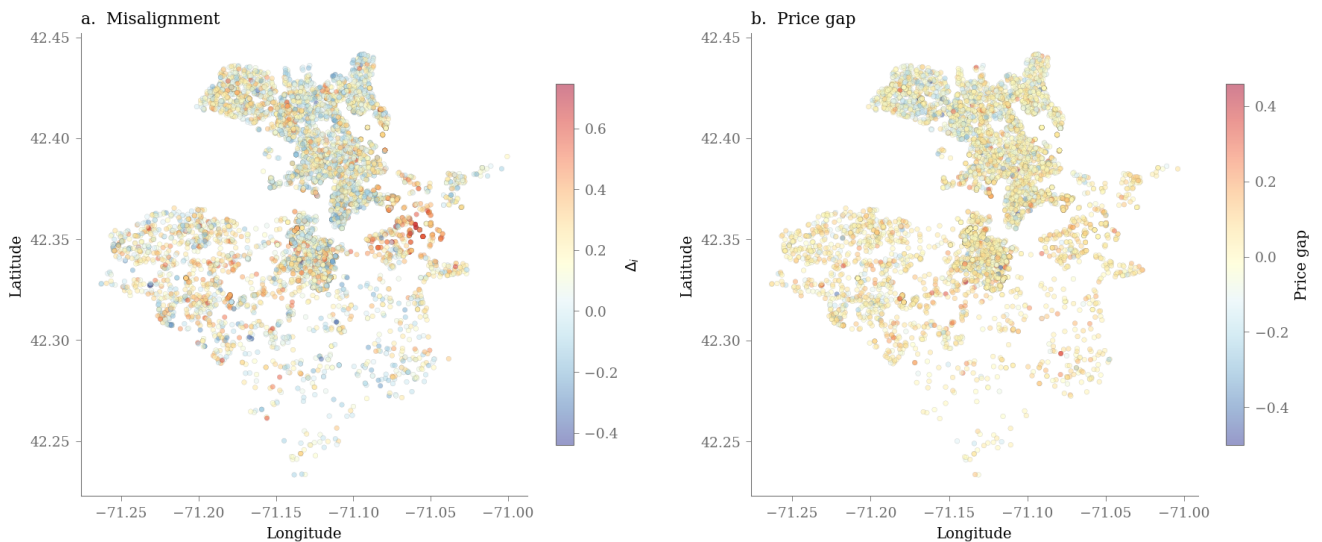


Figure 22: (a) Misalignment is highest in luxury markets. (b) Price gap shows a partially correlated spatial pattern.

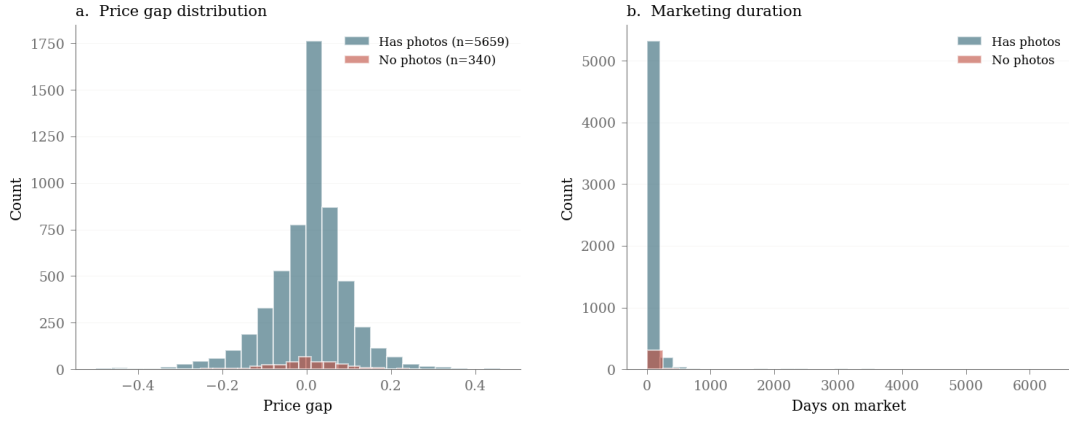


Figure 23: No-photo listings show slightly better outcomes, contradicting the hypothesis that withholding photos is adversarial.

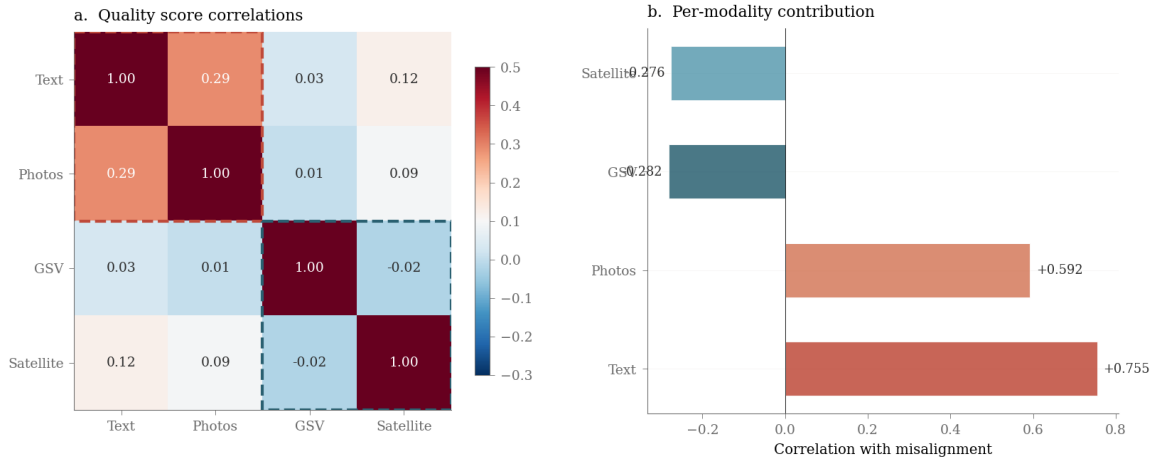


Figure 24: **(a)** Quality score correlations. Dashed boxes: strategic (red), objective (blue). **(b)** Per-modality contribution to Δ_i .

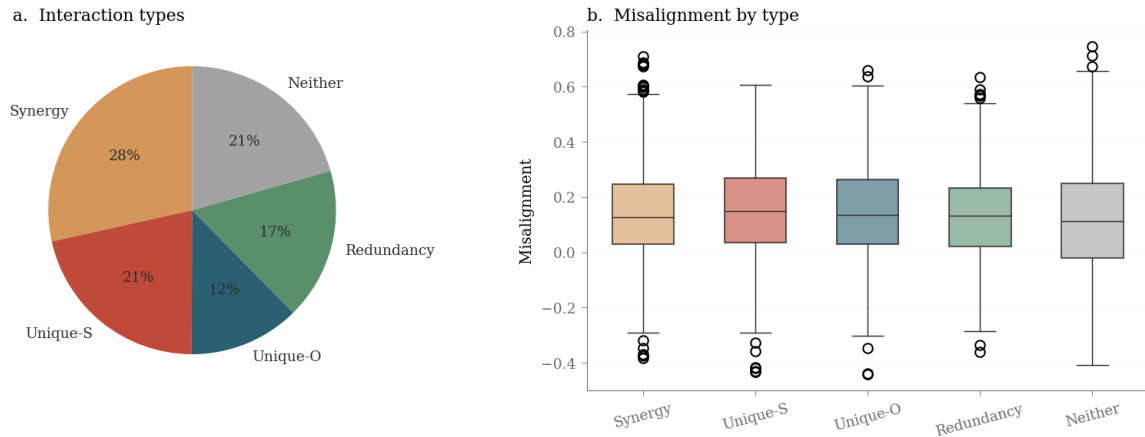


Figure 25: **(a)** Interaction type distribution. **(b)** Misalignment by interaction type.